

Optical Fuses and Isolation using 1D Grating Couplers against QKD Trojan Horse Attacks

Zoe C. M. Davidson^(1, 2), Callum Jones^(2, 3), Nathan Moses⁽²⁾, Siddarth K. Joshi⁽²⁾

⁽¹⁾ Optical and Quantum Centre of Excellence, Research and Commercialisation, BT Group, zoe.davidson@bt.com

⁽²⁾ Quantum Engineering Technologies (QET) Laboratories, University of Bristol, UK,

⁽³⁾ Current affiliation: Department of Physics and Astronomy, University of Exeter, UK.

Abstract *The security of chip-based quantum key distribution (QKD) systems must be secured against high-power laser attacks. We test 1D grating coupler optical fuses as passive countermeasures. We find a heating-induced limit in back-reflections in addition to the damage threshold and discuss implications for QKD. ©2026 The Author(s)*

Introduction

Recent advances in photonic integration now allow chip-based optical platforms that overcome the scalability and reproducibility limits of bulk optics. These integrated approaches are particularly attractive for quantum key distribution (QKD), enabling compact, manufacturable systems suited for large-scale deployment. Ensuring their robustness against side-channel attacks is therefore essential, and this work specifically targets that gap.

Among known QKD vulnerabilities—denial of service (DoS) attacks, detector back-flash [1], high-power laser attacks, basis-selection compromise, and timing manipulation—high-power Trojan Horse attacks (THAs) are of particular concern for integrated chips. Unlike bulk systems, chip-based architectures exhibit strong distributed back-scattering [2, 3], masking component-specific reflections that would otherwise appear as discrete peaks [4]. This makes practical characterisation of side-channel resilience fundamentally different for integrated systems. A promising counter-measure against THAs is the use of optical isolators to suppress back-reflections, coupled with optical fuses (Figure 1): passive components which exhibit a drop in optical transmission above a threshold input power to the device [5, 6]. On-chip components such as grating couplers could act as integrated optical fuses if their optical damage properties are well characterised.

While prior studies have assessed fibre-to-chip couplers [7, 8] and established material damage thresholds for Si and SiN [9–12], crystalline silicon under ultrafast pulses [13], and damage thresholds of edge couplers [14] there remains almost no experimental evidence for damage thresholds in commonly used grating coupling structures. Existing limits on optical power used for QKD security bounds are largely theoretical or assumption-driven, such as in the work of Luca-

marini et al. [15], leaving a critical gap in practical validation.

This work addresses that gap by experimentally evaluating an optical-isolator/optical-fuse configuration and, for the first time, directly characterising the optical damage thresholds of 1D grating couplers under continuous-wave illumination. In addition, we find an optical heating-induced transmission decrease below the damage threshold, demonstrating their potential as integrated optical fuses for chip-based QKD.

Experimental Methods

Photonic integrated chips were fabricated through the CORNERSTONE multi-project wafer service using a silicon-on-insulator (SOI) platform. To test the performance of 1D grating couplers, 1D-1D loopback test structures were used; four identical structures were used on two test chips.

Input and output single mode fibres on a V-groove array (VGA) were aligned to input and output 1D gratings using a 3 dBm continuous-wave (CW) laser diode at 1550 nm (Pritel). The transmission of both 1D gratings was calibrated using this low-power laser.

For high-power tests the laser was amplified to up to 40 dBm with an erbium-doped fibre amplifier EDFA (Keopsys) and transmitted through the 1D-1D loopback structures. The input power was monitored on a photodiode after a 1% splitter and 20 dB attenuator; the output power was measured on a photodiode with a 20 dB attenuator to prevent saturation. Clean fibre connections and a fibre fuse detection system were used to prevent optical damage to instruments.

Results

The relevant parameter for establishing a THA security bound is the return power, i.e. the reflected power detectable off chip. Assuming only the 1D grating coupler transmission T and internal reflectivity R , the return power for an attacker input

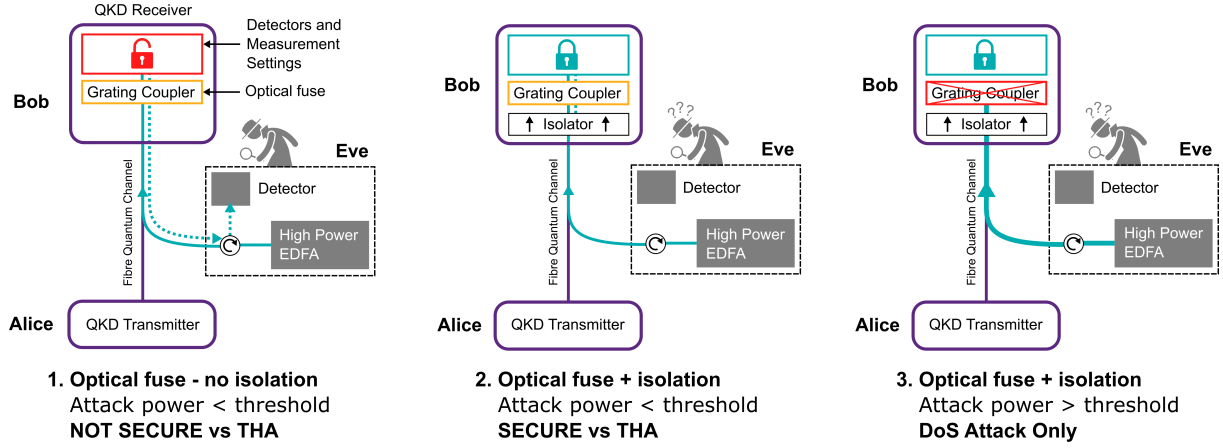


Fig. 1: The Optical Isolator with Fuse Concept. 1. With an optical fuse and no isolation, Eve can use back-reflections to gain information on Bob's measurement bases, i.e. Trojan Horse Attack (THA). 2. Back-reflections are attenuated through the use of an isolator to provide security against THA. 3. The optical fuse can be used in tandem with an isolator to prevent high optical inputs from leaking information. Attacks above the power threshold of the optical fuse instead become denial-of-service (DoS) attacks.

power P_{in} is

$$P_R = RT^2 P_{in}. \quad (1)$$

To set a conservative, coupler-only bound, we take $R = 1$, giving the maximum possible return power $P_{R,max}$. Since T^2 corresponds to the measured two-pass transmission of the loopback structures, any thermally induced reduction in transmission directly reduces $P_{R,max}$.

Optical heating and impact on security bound

As the input power was increased from 31.7 dBm to 40 dBm, the transmission—and therefore $P_{R,max}$ —dropped by several orders of magnitude, but recovered after each cycle. Temperature measurements without stabilisation showed $> 10^\circ\text{C}$ increases at the thermistor, implying significantly higher local heating at the grating coupler as shown in Figure 2a, with a $P_{R,max}$ of approximately 10 mW.

Assuming the measured coupling efficiency of -10 dB per coupler, the absorbed power is $P_{abs} \approx 1$ W for a 10 W incident beam focused to $d \approx 10 \mu\text{m}$ ($A \approx 80 \mu\text{m}^2$). Using the thermal spreading resistance $R_{spread} \approx \frac{1}{4k_{Cu}(d/2)}$ with $k_{Cu} \approx 401 \text{ Wm}^{-1}\text{K}^{-1}$ gives $R_{spread} \approx 125 \text{ K/W}$ and a local temperature rise

$$\Delta T \approx P_{abs} R_{spread} \approx 125 \text{ K}.$$

With the copper-plate temperature at 38°C , the estimated silicon temperature is $T_{Si} \approx 163^\circ\text{C}$, well below the silicon melting point of 1414°C [16–18].

Temperature-controlled tests as seen in Figure 2b (set to 21°C) suppressed the bulk heating but still showed abrupt transmission drops around 32 dBm, attributed to permanent damage discussed later. Although stabilisation shifts the peak in $P_{R,max}$ to higher P_{in} , $P_{R,max}$ eventually decreases again in the high-power regime.

Across all measurements, $P_{R,max}$ exhibits a clear upper limit: heating reduces 1D

grating-coupler transmission before material damage occurs. Thus, the grating coupler naturally limits the return power available to an attacker, acting as an *optical fuse* and enforcing a fundamental upper bound in the THA security model.

Using the QKD security analysis from Ref. [15] and replacing the maximum effective power the attacker can use with our measured 10 mW, we obtain the required optical isolation at the QKD receiver to limit leaked photons per QKD pulse to $\mu_{out} < 10^{-6}$. For consistent comparison we assume the reflectivity of internal chip components $|R| = 50$ dB. Table 1 shows these values (rounded up to the nearest 10 dB) are 30 dB lower than those found using the assumptions on optical damage thresholds in Ref. [15].

Tab. 1: Comparison of total isolation $|I|$ required to limit leaked photons to $\mu_{out} < 10^{-6}$ per QKD pulse between Ref. [15] and the thermally-limited return power from the 1D grating coupler in this work (10 mW). The security analysis from Ref. [15] was used to make this comparison, assuming reflectivity $|R| = 50$ dB. The QKD pulse repetition rate is f_A .

f_A / Hz	$ I $ / dB Ref. [15]	$ I $ / dB This work
10^9	120	90
10^6	150	120
10^3	180	150

Permanent damage - packaged chip testing

Although test chip structures could be repeatedly exposed to 40 dBm CW illumination, a few structures were permanently damaged and visibly burned at 34.6–36.3 dBm. Since this often happened on the second or third high power cycle, we attribute this damage to dust landing and igniting on the grating coupler surface.

As a point of interest we tested a packaged chip to see if there was a difference in outcome when the VGA was fixed to the waveguides using UV cured epoxy resin that is index matched and in a dust-free environment. The epoxy used to fix

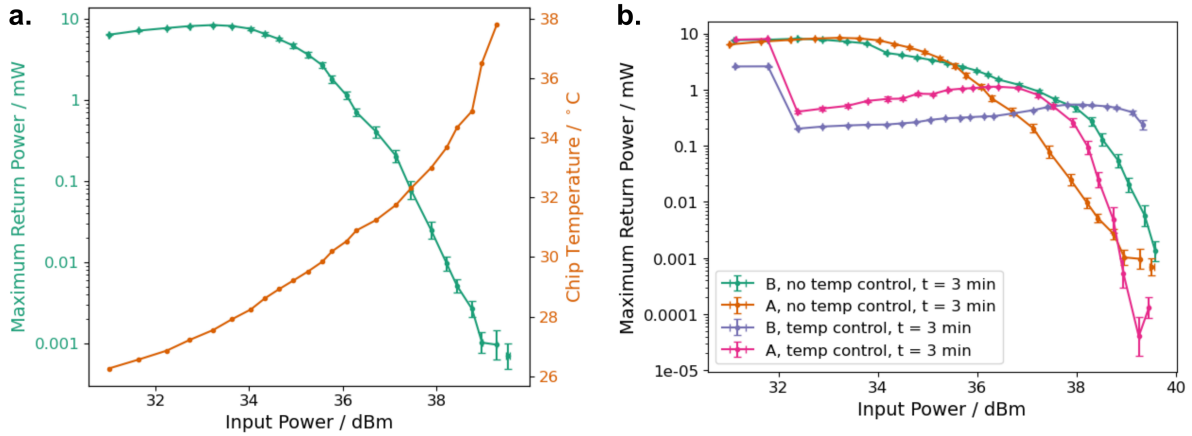


Fig. 2: Effect of temperature control on maximum return power from 1D-1D loopback structures. a. Maximum return power plotted alongside the measured chip temperature, showing a $> 10^{\circ}\text{C}$ temperature increase during high power illumination. b. Two structures A, B tested with and without active temperature stabilisation (setpoint 21°C). With temperature control the peak in maximum return power moves to higher input power but still has an upper bound. Decreases around 32 dB are attributed to permanent damage events.

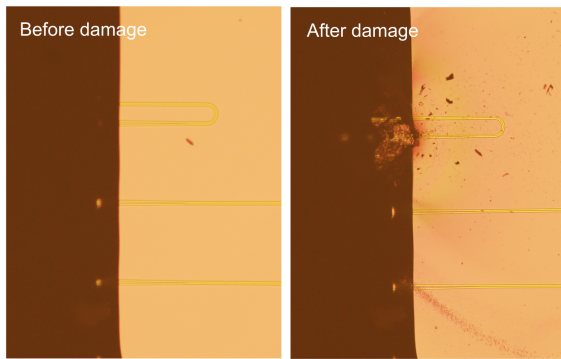


Fig. 3: Microscope images of 1D grating loopback structure input on the packaged chip before and after optical damage.

the VGA to the chip was Norland Optical Adhesive 81 (NOA-81) which is temperature rated for -150°C to 125°C . The low power (3 dBm) coupling of the packaged chip showed an insertion loss of -17 dBm, this is approximately 2 dB improvement in insertion loss of the best structure tested in the previous measurements.

The packaged chip was then tested under high power starting at 31.7 dBm and increasing to 35.8 dBm where some transmission was lost, before total loss of transmission and visible damage in the epoxy was seen at 38.3 dBm, seen in Figure 3.

As calculated previously the temperature generated by the beam spot size on the Si was estimated to be in the region of 163°C with a free space coupled VGA. This exceeds the temperature bound of the NOA-81, and thus damage occurred. This raises additional implications for high power laser attacks on packaged chips, depending on the temperature rating and cure process of the epoxy which affixes the fiber to the chip.

Conclusions

We have experimentally investigated 1D grating couplers operated as passive optical fuses against high-power Trojan Horse Attacks in chip-based QKD systems. By directly measuring transmission, return power, and damage behaviour under CW illumination, we show that thermally induced self-attenuation imposes a fundamental upper bound on the attacker-accessible return power before irreversible material damage occurs. This heating-driven optical fuse effect converts high-power laser eavesdropping attempts into denial-of-service events. Comparing our measurements with established THA security analyses [15], we find a potential 30 dB reduction in optical isolation required on a chip-based QKD receiver device.

Our results provide the first experimental validation of damage and pre-damage behaviour in grating coupling structures, closing an important gap between theoretical security models and practical integrated implementations. We further demonstrate that packaging choices play a critical role: index-matched epoxies can fail at temperatures well below silicon damage thresholds, introducing new attack vectors that impact alignment and system availability rather than information leakage.

These findings emphasise that side-channel security in integrated QKD systems is governed not only by optical design but by coupled thermal, material, and packaging constraints. Incorporating fuse-like behaviour into coupling structures, alongside appropriate isolation and high-temperature packaging strategies, offers a promising route to robust, passive mitigation of high-power laser attacks. Future work will extend this analysis to alternative coupling geometries, pulsed attack regimes, and co-designed security bounds that explicitly include thermo-optical effects in integrated photonic platforms.

Acknowledgements

This work was supported by QAssure, an Innovate UK-funded project (no.10102791), and equipment from the UK EPSRC grant QuPIC (EP/N015126/1). The packaged chip was provided by the PIC Bootcamp, an Innovate UK-funded project (no.10074923). The authors would like to thank Marcus Clark for assistance with the experimental setup, Friederike Jöhlinger for valuable discussions and designing an earlier version of the setup, and Cathy White for excellent leadership of the QAssure project.

References

- [1] A. Meda, I. P. Degiovanni, A. Tosi, Z. Yuan, G. Brida, and M. Genovese, "Quantifying backflash radiation to prevent zero-error attacks in quantum key distribution", *Light: Science & Applications*, vol. 6, e16261, 2017. DOI: 10.1038/lsa.2016.261. [Online]. Available: <https://doi.org/10.1038/lsa.2016.261>.
- [2] H. Tan, W. Li, L. Zhang, K. Wei, and F. Xu, "Chip-based quantum key distribution against trojan-horse attack", *Physical Review Applied*, vol. 15, no. 6, p. 064038, 2021. DOI: 10.1103/PhysRevApplied.15.064038. [Online]. Available: <https://doi.org/10.1103/PhysRevApplied.15.064038>.
- [3] F. Jöhlinger et al., *Physical security of chip-based quantum key distribution devices*, 2024. arXiv: 2408.16835 [quant-ph]. [Online]. Available: <https://arxiv.org/abs/2408.16835>.
- [4] N. Jain, E. Anisimova, I. Khan, V. Makarov, C. Marquardt, and G. Leuchs, "Trojan-horse attacks threaten the security of practical quantum cryptography", *New Journal of Physics*, vol. 16, p. 123030, 2014. DOI: 10.1088/1367-2630/16/12/123030. [Online]. Available: <https://doi.org/10.1088/1367-2630/16/12/123030>.
- [5] E. Borisova, A. Ponosova, B. Galagan, V. Koltashev, N. R. A. Rutyunyan, and V. Makarov, "Optical fuse as a countermeasure against light injection attacks on quantum key distribution systems", in *2024 International Conference Laser Optics (ICLO)*, IEEE, Jul. 2024, pp. 103–103. DOI: 10.1109/iclo59702.2024.10623944. [Online]. Available: <http://dx.doi.org/10.1109/ICLO59702.2024.10623944>.
- [6] M. Chen et al., *Optical fuse based on the photorefractive effect for defending the light-injection attacks of quantum key distribution*, 2025. DOI: 10.48550/ARXIV.2512.10205. [Online]. Available: <https://arxiv.org/abs/2512.10205>.
- [7] X. Mu, S. Wu, L. Cheng, and H. Y. Fu, "Edge couplers in silicon photonic integrated circuits: A review", *Applied Sciences*, vol. 10, no. 4, p. 1538, 2020. DOI: 10.3390/app10041538. [Online]. Available: <https://doi.org/10.3390/app10041538>.
- [8] L. Cheng, S. Mao, Z. Li, Y. Han, and H. Y. Fu, "Grating couplers on silicon photonics: Design principles, emerging trends and practical issues", *Micromachines*, vol. 11, no. 7, p. 666, 2020. DOI: 10.3390/mi11070666. [Online]. Available: <https://doi.org/10.3390/mi11070666>.
- [9] M. L. V. Gayler, "Melting point of high-purity silicon", *Nature*, vol. 142, p. 478, 3593 1938. DOI: 10.1038/142478a0. [Online]. Available: <https://doi.org/10.1038/142478a0>.
- [10] A. E. Kaloyeros, Y. Pan, J. Goff, and B. Arkles, "Review—silicon nitride and silicon nitride-rich thin film technologies: State-of-the-art processing technologies, properties, and applications", *ECS Journal of Solid State Science and Technology*, vol. 9, no. 6, p. 063006, Aug. 2020. DOI: 10.1149/2162-8777/aba447. [Online]. Available: <https://doi.org/10.1149/2162-8777/aba447>.
- [11] S. Singer et al., "Nonlinear loss and damage threshold in silicon photonic waveguides: Modelling and experimental verification", in *2022 Conference on Lasers and Electro-Optics (CLEO)*, 2022, pp. 1–2.
- [12] S. Tan et al., "Silicon nitride waveguide as a power delivery component for on-chip dielectric laser accelerators", *Opt. Lett.*, vol. 44, no. 2, pp. 335–338, Jan. 2019. DOI: 10.1364/OL.44.000335. [Online]. Available: <https://opg.optica.org/ol/abstract.cfm?URI=ol-44-2-335>.
- [13] B. Cowan, "Optical damage threshold of silicon for ultrafast infrared pulses", in *AIP Conference Proceedings*, vol. 877, American Institute of Physics, Nov. 2006, pp. 837–843. DOI: 10.1063/1.2409223. [Online]. Available: <https://doi.org/10.1063/1.2409223>.
- [14] F. Jöhlinger, "Security of chip-scale quantum key distribution", Ph.D. dissertation, University of Bristol, 2023.
- [15] M. Lucamarini, I. Choi, M. B. Ward, J. F. Dynes, Z. L. Yuan, and A. J. Shields, "Practical security bounds against the trojan-horse attack in quantum key distribution", *Phys. Rev. X*, vol. 5, p. 031030, 3 Sep. 2015. DOI: 10.1103/PhysRevX.5.031030. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevX.5.031030>.
- [16] Y. S. Muzychka, J. R. Culham, and M. M. Yovanovich, "Thermal spreading resistance in compound and orthotropic systems", *International Journal of Thermal Sciences*, vol. 43, pp. 385–394, 2004, Provides spreading resistance correlations and multilayer solutions; used here for the $1/(4ka)$ approximation.
- [17] E. Toolbox. "Thermal conductivity of metals and alloys — copper". Typical copper thermal conductivity values ($\approx 390\text{--}401\text{ W m}^{-1}\text{K}^{-1}$); accessed 2025-11-03.
- [18] R. S. of Chemistry. "Silicon — element information, properties and uses". Provides thermal properties and melting point of silicon; accessed 2025-11-03.